

Todo espacio L^2 es un espacio de Hilbert

Teorema 1. Sea (X, Σ, μ) un espacio de medida

$\implies L^2 := \mathcal{L}^2 / \sim$ es un espacio de Hilbert con el producto interno $\langle \cdot, \cdot \rangle$ dado por

$\forall f, g \in L^2 : \langle f, g \rangle = \int_X f \cdot \bar{g} d\mu$ donde $f \sim g \iff f = g$ c.t.p.

Demostración: Veamos primero que $\langle \cdot, \cdot \rangle$ es un producto hermítico en L^2 :

(i) Sesquilinealidad: $\forall f, g, h \in L^2 : \forall \alpha, \beta \in \mathbb{C} :$

$$\begin{aligned} \langle \alpha f + \beta h, g \rangle &= \int_X (\alpha f + \beta h) \cdot \bar{g} d\mu \stackrel{\text{lineal}}{=} \alpha \int_X f \cdot \bar{g} d\mu + \beta \int_X h \cdot \bar{g} d\mu = \alpha \langle f, g \rangle + \beta \langle h, g \rangle \\ \langle f, \alpha g + \beta h \rangle &= \int_X f \cdot \overline{(\alpha g + \beta h)} d\mu \stackrel{\text{lineal}}{=} \bar{\alpha} \int_X f \cdot \bar{g} d\mu + \bar{\beta} \int_X f \cdot \bar{h} d\mu = \bar{\alpha} \langle f, g \rangle + \bar{\beta} \langle f, h \rangle \end{aligned}$$

(ii) Simetría conjugada: $\forall f, g \in L^2 : \langle f, g \rangle = \int_X f \cdot \bar{g} d\mu = \overline{\int_X g \cdot \bar{f} d\mu} = \overline{\langle g, f \rangle}.$

(iii) Definida positiva: $\forall f \in L^2 : \langle f, f \rangle = \int_X |f|^2 d\mu \geq 0$ y $\langle f, f \rangle = 0 \iff f = 0$ c.t.p.

Solo falta ver que L^2 es de Banach con la norma inducida por el producto interno:

$$\forall f \in L^2 : \|f\| = \sqrt{\langle f, f \rangle} = \left(\int_X f \cdot \bar{f} d\mu \right)^{1/2} = \left(\int_X |f|^2 d\mu \right)^{1/2} = \|f\|_2.$$

Esta es exactamente la norma 2-ésima y, por el `teo-esp-lp-banach/??`, sabemos que L^2 es un espacio de Banach con ella. ■

Referenciado en

- `teo-l2-unico-esp-hilbert`

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